G1 Digital-Twin Payload Experiment

Methods, spill model, governance algorithm and full campaign results — supporting material for Section VII.H

Simulation campaign 2026-07-04 · Unitree G1 · Gazebo (ROS 2 Humble) · 99 simulation runs, 82 valid formal-campaign runs across 11 arms · rev. 2 (four-layer architecture + runtime certification ratio)

1. Digital twin: Gazebo replica of the laboratory

The physical robot is a Unitree G1 "Air" humanoid, a consumer unit whose only external interface is a single WebRTC session through the vendor app; the production navigation stack drives it by tapping that session over USB. For the twin we reuse **the exact same stack, unmodified**, and substitute only its interface object, so any behaviour observed in simulation is produced by the identical code deployed on hardware.

The simulator runs in a Docker container (ROS 2 Humble Desktop, Gazebo Classic, noVNC). The world, **lab.world**, is generated programmatically from the robot's own SLAM maps of the real laboratory: wall occupancy cells become 2.2 m box obstacles in the robot's coordinate frame, so the twin's geometry, waypoints (A, B, C) and the 0.8 m doorway coincide with the real deployment.

1.1 Robot model and physics

The robot is spawned from a URDF whose collision geometry is a single box matching the G1's real envelope (0.44 m shoulder width × 1.10 m tall, capped below the simulated LiDAR at z ≈ 1.20 m so the sensor does not see itself). High mass (60 kg) and large diagonal inertia keep the base upright under the velocity-controlled gait; ground friction is zero so the planar-move plugin drives it as a velocity-controlled puck while wall contacts remain solid. Locomotion is commanded as gait velocity through gazebo_ros_planar_move (/cmd_vel, /odom back); a 360-ray LiDAR at 10 Hz publishes /scan. The doorway was re-carved to the measured physical opening after the collision model was corrected: with the original carve the clear gap was 0.42 m and the realistic body physically could not pass — an artefact caught by consistency checking before any campaign data was taken.

1.2 The adapter: same stack, simulated interface

- **Velocity commands** {lx, ly, rx, ry} → /cmd_vel, using the physical robot's measured stick-to-velocity mapping (deadzone subtracted, not clipped; ly = 0.40 → 0.30 m/s), so simulated and real dynamics coincide.
- **Pose** ← /odom (map and odometry frames coincide in simulation: no relocalisation jitter).
- **LiDAR** ← /scan, projected into a flat world-frame point cloud identical in shape to the real WebView decoder output.

Camera perception is disabled in the twin; the LiDAR-driven planner and the governance layer are exercised in full. Data runs are always headless.

2. Spill model (payload hazard)

Spilling is modelled physically, not as an arbitrary speed threshold. The water surface at the cup rim is a **damped second-order oscillator** — the first sloshing mode of an open cylinder — excited by the horizontal acceleration of the cup:

$$\omega_0 = \mathrm{sqrt}(\ g \cdot k_1 \cdot \tanh(k_1 \cdot h)\ ),\quad k_1 = 1.841 / R \qquad (\approx 21 \text{ rad/s for } R = 4 \text{ cm},\ h = 8 \text{ cm})$$

$$\ddot{\eta} = -2 \cdot \zeta \cdot \omega_0 \cdot \dot{\eta} - \omega_0^2 \cdot (\eta + R \cdot a\_cup/g) \qquad \text{per body axis},\quad \zeta = 0.06$$

The cup is held ~0.25 m ahead of the base, so a_cup = a_base + α×r + ω×(ω×r): fast turning adds a lever term. Bipedal gait injects acceleration noise growing with speed ($\approx v^2$; 1.4 Hz component plus broadband) — **walking fast agitates the cup by itself**. A sharp deceleration while forward motion is still commanded (an external impact, not a commanded stop) adds a direct impulse. Commanded acceleration is low-pass filtered ($\tau = 0.25$ s) because the real G1 ramps velocity smoothly whereas the planar-move plugin steps it instantaneously.

Spills are drawn from a **non-homogeneous Poisson process** on how far the surface exceeds the freeboard (12 mm below the rim):

$$\lambda(t) = LID \cdot \lambda_0 \cdot \max(0,\ |\eta|/freeboard - 1)^2 ,\quad \lambda_0 = 40\ s^{-1} ,\quad LID = 1.0 \text{ (open)} / 0.25 \text{ (sealed)}$$

Each spill loses water (freeboard grows) and dissipates slosh energy. The model is **observational only**: it never alters physics or control. Calibration against logged trajectories yields ≈ 2 spills per 80 s of sustained 0.30 m/s cruising and ≈ 0 at the governed 0.18 m/s ceiling, consistent with the jerk-dominated design of the 2-D thesis study. For the physical robot the same event channel accepts a **manual ground-truth mark** (a keypress or an empty ROS message per observed spill), timestamped into the identical dataset field.

3. Governance algorithm (DCE runtime)

The governance layer is the configuration-first meta-reasoner (Meta-Reasoner 2.0) fed at 2 Hz through a bridge that packages the stack's per-tick telemetry as shared-experience readings. Each tick the reasoner evaluates the candidate analogies (Efficient_Nav / Cautious_Nav) against their QoE boundaries — analogy-level DST intervals, semantic-region memory and direction, tension with attention exponents, fulfillment, required-meta, uncertainty and hard-veto gates — and returns KEEP / SWITCH / FALLBACK / HELP / INSUFFICIENT plus the active analogy. The bridge maps the decision to a forward-speed ceiling (Efficient = uncapped ≈ 0.30 m/s, Cautious = 0.18 m/s, FALLBACK = 0.14 m/s, HELP = 0).

Meta-parameter	Source (per-tick telemetry)	Role
safety	clearance = c0 / 1.5 m (LiDAR forward free space)	QoE regions + hard veto
progression	goal-approach rate / 0.30 m/s (2 s window)	required-meta threshold
mobility	achieved / commanded speed (1.2 s window)	resistance channel, hard veto
fragility	payload state: 1.0 intact, drops per spill	hard veto (payload task)
reliability / uncertainty	sensing monitor + few-shot historical margin	DST belief interval

Readings are median-smoothed (~2.5 s) and each reading's DST uncertainty adds the historical dispersion of its own metric (few-shot margin). Switches are hysteretic; an experience-escalation layer aborts on sustained HELP or a stagnant high-FALLBACK window.

3.1 Realising the four-layer belief architecture of thesis Chapter 5

Chapter 5 of the thesis specifies three DST belief layers plus a deployment-level Meta-Attention layer, each addressing a distinct failure mode at its own timescale. Table 2 maps each layer to its realisation on the G1. Layers 1-2 run in every governed experiment; Layers 3-4 are implemented and validated but opt-in (they were disabled during the reported campaign so that the published arms remain exactly reproducible).

Thesis layer	Timescale	G1 realisation	Status
L1 — path-hazard plausibility (Pl_H): collapse transfer, force conservative	sensor frame	Per-tick analogy-level DST on the safety reading with QoE regions and hard veto inside the reasoner (strips certification in ~2 ticks; measured ≈4 s correction of a task-blind prior), over the stack's own hard safety guards	covered (equivalent mechanism)
L2 — analogy transfer trust (Pl_analogy): outcome evidence per run	run	Per-analogy Dempster–Shafer belief {match, mismatch, Θ} persisted between runs (G1_M2_STATE); spill → mismatch mass, clean run → match mass; Pl < 0.70 blends the ceiling toward conservative, Pl < 0.45 vetoes deployment and corrects the start-up prior	implemented
L3 — environment relevance (Pl_env): relax / escalate spatial assumptions	run (continuous)	EMA of the median clearance relative to the active analogy's expected boundary, clamped to [0.85, 1.15]; scales profile ceilings only (never FALLBACK/HELP; bounds 0.22-0.40). Validated: open environment raises the Cautious ceiling 0.28 → 0.32, narrow lowers it → 0.26	implemented (opt-in G1_M2_L3)
L4 — Meta-Attention: chronic low fulfilment despite high trust → profile mismatch	deployment (run window)	Per-run mean fulfilment history persisted with the L2 state; fulfilment < 0.45 over 3 consecutive runs with Pl ≥ 0.45 under the same profile → start-up re-selection of the alternative profile. The experience-escalation abort (sustained HELP) covers the within-run part of this timescale	implemented (opt-in G1_M2_L4)

Table 2. Thesis Ch. 5 four-layer architecture mapped to the G1 runtime. All four layers live in the bridge; the meta-reasoner itself is untouched.

Two details differ from the 2-D study, both additive. First, the **fragility channel acts within the run** (one spill → caution ≈ 0.52, two → FALLBACK band ≈ 0.31, three or more → dangerous < 0.12 and a hard-veto HELP; ~25 s recovery), which the 2-D study could not do because its spill evidence arrived only per run — the tick-level DST of the runtime makes the intra-run channel possible. Second, spill evidence is attributed to the analogy **governing at the moment of the spill**, not to the whole run, sharpening the Layer-2 credit assignment.

3.2 Runtime certification ratio (ρ_DCA)

The paper's robustness certificate is instantiated as a runtime quantity and logged with every governance decision:

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ρ̂= max(0, Δmargin) / ( task_uncertainty_gap(winner) + 0.5·θ_mismatch(winner) + 0.02 )
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where Δmargin is the stable-fulfilment gap between the two best deployable candidates (or the distance to the certification threshold when only one is deployable), task_uncertainty_gap is the DST-measured calibration budget (the θ_QoE proxy), and θ_mismatch is the L2 distance between normalised task and analogy attention vectors (computed from configuration; B constants set to 1 and ξ_rep declared unmeasured at runtime). Validation reproduces the paper's regimes empirically: $\hat{\rho} = 1.50$ in an open corridor with clean readings (stable) versus $\hat{\rho} = 0.58$ under degraded, oscillating readings (non-certifiable). The value is recorded per tick (meta2_rho) so certification traces analogous to Table V of the manuscript can be produced from any run.

4. Campaign protocol

99 simulation runs were executed in total: 82 valid formal-campaign runs across 11 governance arms (8 per arm; 6 for sealed-lid arms), plus development and validation runs (adapter bring-up, collision-physics correction, smoke tests) and one run excluded by audit (governance silently disabled by a mid-campaign code edit — detected by a per-tick telemetry check and discarded). Every run resets Gazebo to a fresh world with the robot at waypoint A and executes an autonomous A→B traverse headless; arms alternate to share machine-load drift; datasets are tagged env=sim with a per-arm identifier. Statistics follow the thesis protocol: mean ± 95% CI ($1.96 \cdot s/\sqrt{n}$).

5. Results

Table 3 reports all eleven arms. Risk is the fraction of run time with the liquid surface above 70% of freeboard; HELP counts governed aborts (sustained empty deployable set); times average reached runs only.

Arm	Governance	Reached	Time (s)	Spills/run	Risk (%)
M0 baseline	none (0.30 m/s)	8/8	83.4 ± 5.2	0.50 ± 0.37	8.1 ± 1.4
M1 fixed-conservative	none (0.18 m/s cap)	8/8	110.8 ± 2.4	0.00	0.8 ± 0.2
Single-candidate governed	DCE, one analogy	5/8 (3 HELP)	106.2 ± 5.1	0.12 ± 0.24	0.5 ± 0.2
M2 payload (proposed)	DCE, payload task	8/8	107.9 ± 4.2	0.00	0.4 ± 0.2
M2 task-blind prior	DCE, lab-QoE config	8/8	108.2 ± 3.5	0.00	0.7 ± 0.2
M2 task-blind (twin-cal.)	DCE, twin-QoE config	7/8 (1 HELP)	106.6 ± 5.2	0.12 ± 0.24	0.8 ± 0.6
M2 Wrong-DST	DCE + Layer 2, lab-QoE	8/8	110.6 ± 0.9	0.00	0.7 ± 0.3
M2 Wrong-DST (twin-cal.)	DCE + Layer 2, twin-QoE	8/8	108.4 ± 4.4	0.00	0.6 ± 0.2
M0 sealed lid	none	6/6	88.1 ± 4.9	0.17 ± 0.33	10.6 ± 2.9
M2 sealed lid	DCE, lab-QoE config	4/6 (2 HELP)	111.5 ± 7.3	0.00	1.0 ± 0.4
M2 sealed lid (twin-cal.)	DCE, twin-QoE config	5/6 (1 HELP)	102.5 ± 2.2	0.17 ± 0.33	0.7 ± 0.5

Table 3. Full campaign results (82 valid runs). Lab-QoE = QoE boundaries calibrated on the physical laboratory; twin-QoE = re-calibrated from measured twin distributions.

Figure 1. Safety-time trade-off. The ungoverned baseline is fast but spills on half of its runs; every governed arm collapses into a zero/near-zero-spill cluster at a 25-35% time cost. The tightness of the cluster is itself the finding: governance outcome is robust to which prior is loaded.

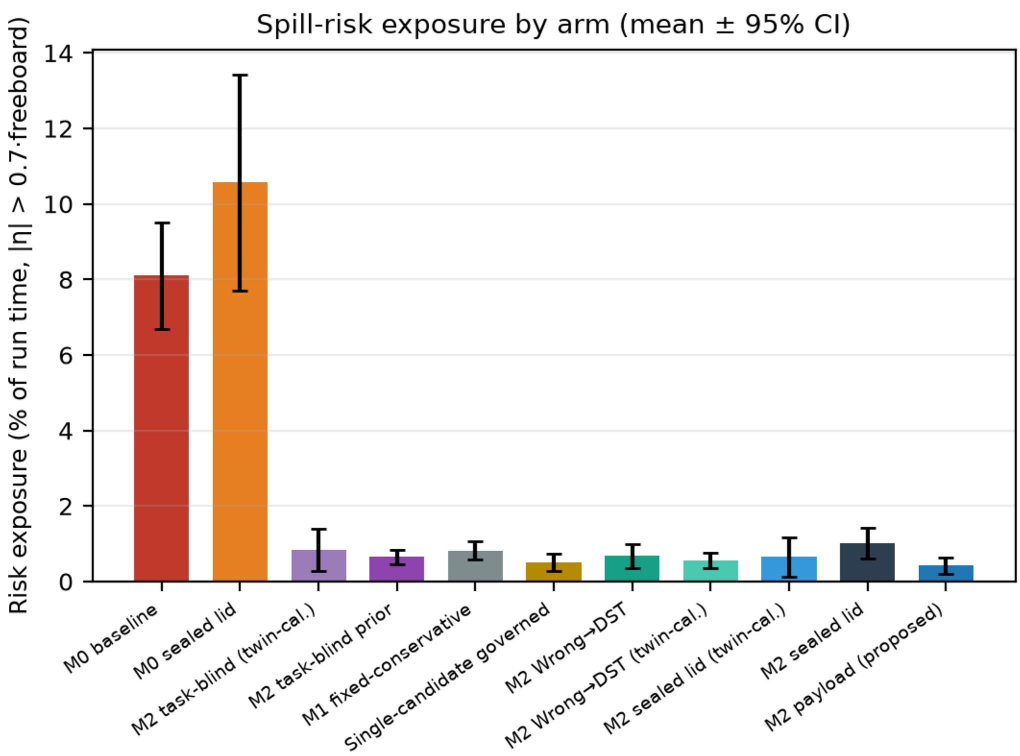

Figure 2. Continuous risk exposure by arm. The ungoverned arms (open and sealed lid) spend an order of magnitude more time near the spill threshold than any governed arm; the sealed lid reduces realised spills but not the exposure, confirming its gain is passive physics.

Finding 1 — deployability governance converts spills into bounded time cost (Fig. 1): M0 spills 0.50/run at 8.1% exposure; M2 payload governance eliminates spills (0/8; 0.4%) at +29% time with 8/8 completion. **Finding 2 — HELP semantics**: with a single-analogy candidate set, a mobility hard veto while pressing the corridor pinch empties the deployable set; sustained HELP aborts the run (3/8). This is the specified conservative non-selection behaviour, not a failure.

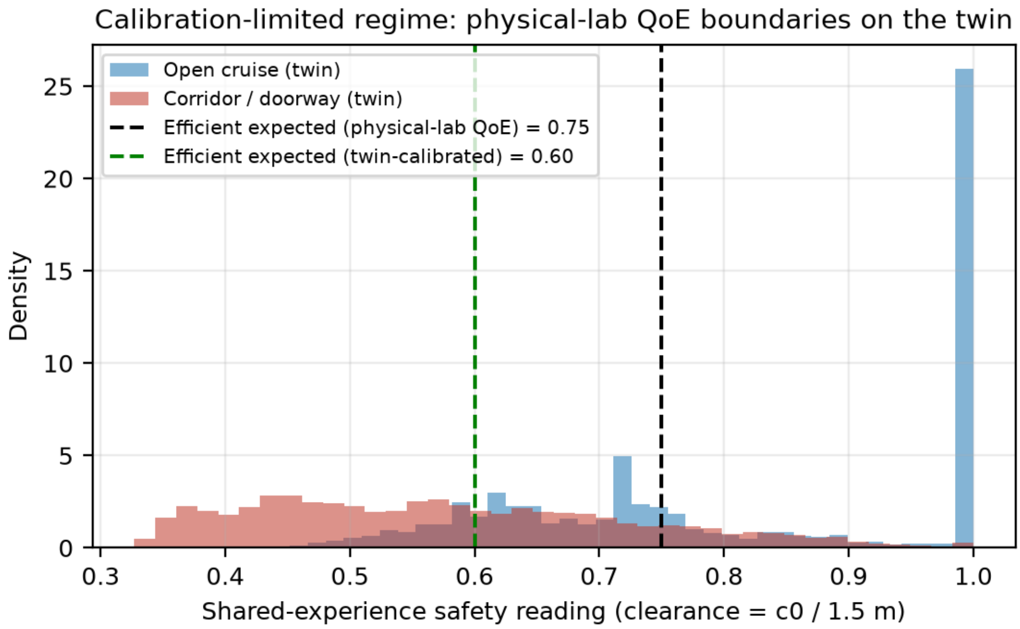

Figure 3. The calibration-limited regime, observed live. The physical-lab QoE boundary for the efficient capability (0.75) sits above the twin's cruise clearance distribution: the capability is never certifiable on the twin until the boundary is re-calibrated from measured distributions (0.60), per the interface-refinement pattern.

Finding 3 — θ_QoE in deployment (Fig. 3): the twin's clearance distribution is systematically narrower than the physical laboratory's; running the lab calibration unchanged leaves Efficient_Nav non-certifiable for entire runs. Re-calibration restored certification in open stretches without touching the reasoner or the physical-robot configuration.

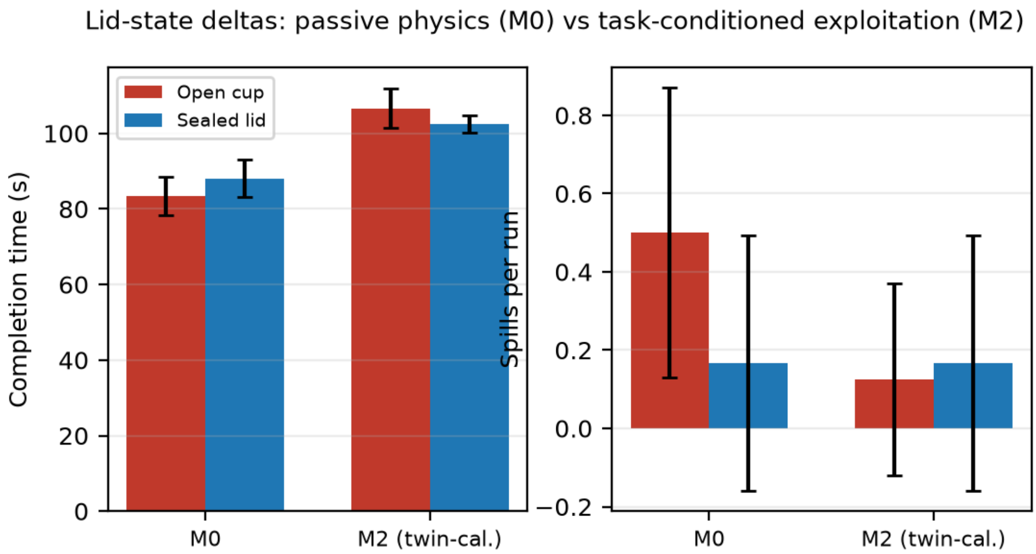

Figure 4. Lid-state deltas. The ungoverned baseline improves only through physics (fewer realised spills, identical exposure); the twin-calibrated governed system also converts the sealed lid into speed (102.5 ± 2.2 s vs 111.5 ± 7.3 s), a policy-level exploitation unavailable to fixed baselines.

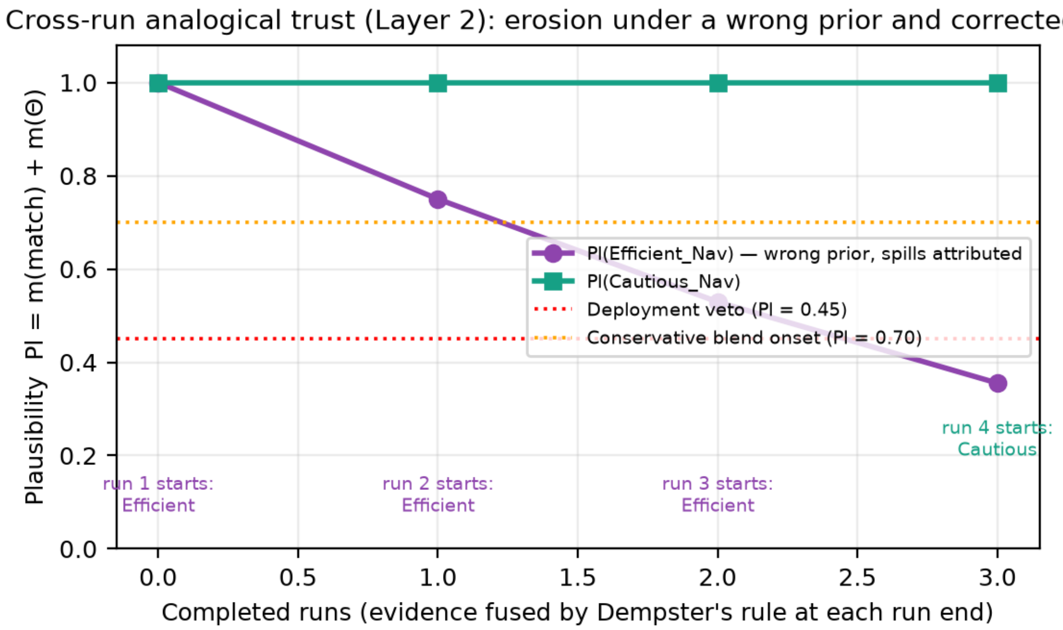

Figure 5. Cross-run analogical trust (Layer 2), mechanistic validation. Under a forced wrong prior with spills attributed to it, plausibility degrades 1.00 → 0.75 → 0.53 → 0.36 by Dempster fusion; crossing the blend threshold (0.70) progressively caps the analogy's speed, and crossing the deployment veto (0.45) makes run 4 start pre-corrected in the conservative analogy.

Finding 5 — temporal scales separate as designed (Fig. 5): in the twin campaign the Wrong-DST arms show flat, spill-free sequences because per-tick shared-experience evidence (the Layer-1 timescale) corrects the task-blind prior within ≈ 4 s of each run — before any spill can occur — so the run-level trust layer receives no failure signal. The mechanism itself is validated in the controlled trace of Fig. 5. Its operational regime is the physical laboratory, where open-floor cruising genuinely certifies the fast capability and gait-induced spills are invisible to per-tick channels; the twenty-run physical campaign exercises it.

Fidelity to theory. Four of the five claims instantiated by the campaign are confirmed with statistics (Findings 1-4); the fifth is confirmed mechanistically with its operational regime correctly predicted to lie on the physical robot. Two artefacts were caught and corrected by consistency checking before affecting conclusions: a run with governance silently disabled (excluded) and a single-candidate arm originally mislabelled as the M1 baseline (retained as the HELP-semantics arm; a true fixed-cap M1 was added).

Datasets, configurations, the aggregation script and figure-generation code are versioned with the robot stack (commits up to the 2026-07-04 campaign; four-layer architecture and runtime certification ratio added in rev. 2).